Qiskit Fall Fest 2026

Challenge 4: Quantum Correlation Switch

Problem Statement - In the challenge 3, how a quantum circuit can produce correlated outcomes is investigated. Now, design a circuit that switches between correlated and opposite measurement outcomes. Your goal is to create two different two-qubit behaviors using only a small number of quantum gates.

Your Task - Using Qiskit:

- **Part A – Correlated Outcomes**
  Design a two-qubit circuit in which the measurement results are predominantly: 00 and 11.
  Run the circuit for **100 shots** and record the results.
- **Part B – Opposite Outcomes**
  Modify your circuit so that the measurement results are predominantly: 01 and 10
  Run the modified circuit for **100 shots** and record the results.
- **Part C – Compare**
  Compare the two circuits using histograms.

Explain:

- Which outcomes are dominant in Part A?
- Which outcomes are dominant in Part B?
- What changed between the two circuits?
- Why did the measurement relationship change?

Design Constraints - Your circuit must:

- Use exactly **2 qubits**.
- Measure both qubits.
- Use no more than **4 quantum gates before measurement**.
- Use only gates introduced during the workshop.
- Be created and executed using Qiskit.
- Use **100 shots** for each experiment.

Think Before You Run - Before executing your circuits, predict: Which outcomes should dominate in Part A? Which outcomes should dominate in Part B? Will all four possible outcomes appear? Why might some outcomes be much more frequent than others?

Submission - Your notebook must include:

- Circuit for Part A.
- Circuit for Part B.
- Predictions for both circuits.
- Measurement results for 100 shots.
- Histograms for both circuits.
- A comparison of the two experiments.
- A short explanation of how the circuit produces the different correlations.